14. Pro-Code Extension Using Custom Functions and Zoho Catalyst

Aim:

To study custom-coded extensions in the Gym Membership application using existing Deluge functions and outline how a secured Catalyst service could support more specialized processing.

Definition:

Pro-code extension adds programmed logic beyond visual configuration. A Deluge custom function provides reusable application behavior. Catalyst can host backend functions and HTTP-facing services. Connecting the two requires an authenticated interface, validated inputs, controlled permissions and explicit failure handling.

Existing Custom Functions:

Function	Behavior in the Export
Refresh_Report_Counts	Loops through enrollment records and calculates representative daily and weekly labels for report views.
Send_Renewal_Reminders	Selects members whose end date falls within a seven-day window and whose email is present; sends renewal messages with shared rules content.
Open Members for Start Date action	Opens All Members List filtered by the selected preferred starting date.

Integration Points:

Refresh_Report_Counts is invoked after enrollment add/edit success and by a daily schedule. The renewal function is referenced by the Command Center button. The calendar action opens a filtered report. These are configured links; screenshots of the controls do not establish that every invocation or email delivery succeeded.

Custom-function Logic:

For renewal reminders, set start_date to the current date and end_date to start_date + 7 days. Fetch records with start_date <= Membership_End_Date < end_date and a non-null email. Skip blank trimmed addresses, send the message and count submitted send operations. A send counter is not proof of recipient delivery.

Procedure:

1. Open the Functions and Workflows areas and inspect the existing custom functions and their callers.

2. Review field names, date boundaries, null handling and output messages before changing code.

3. Test report-count refresh in a development copy with several members sharing a date and week. Verify labels against source records.

4. Test renewal selection using today, today + 6 and today + 7 days, blank email values and an empty selection. Use controlled recipients.

5. Prevent repeated reminder runs from sending duplicates when the business policy requires this, for example by adding a reminder-history record and checking its state. This is a proposed enhancement.

6. For Catalyst exploration, define one suitable backend task such as a computationally intensive membership analysis using minimized input data.

7. Create a development backend function and define the request/response contract. Configure authentication and authorization before connecting it to Creator.

8. Call the endpoint from Deluge using invokeurl and a supported connection/authentication method. Validate the response and handle timeouts, non-success status and malformed data.

9. Test allowed and denied requests, duplicate requests and backend failure. Retain a traceable request identifier without logging sensitive member data.

Proposed API Contract:

Item	Design
Request	POST to a configured HTTPS endpoint; send only the fields needed for the selected task.
Identity	Use a non-secret request identifier and a member record reference as needed.
Authorization	Validate caller identity and allowed operations on the server; do not trust a supplied member ID by itself.
Response	Return a status, request identifier and validated result fields in a defined JSON structure.
Failure	Return a controlled error; preserve local data and show or log an actionable message.

Verification Cases:

Case	Expected Outcome
Renewal ending today	Included when an email is present.
Renewal ending in six days	Included.
Renewal ending in seven days	Excluded by the existing upper-exclusive criterion.
Blank or missing email	No email send attempted for that record.
Repeated reminder invocation	Current function may resend; a history/idempotency enhancement is needed for duplicate prevention.
Backend rejects credentials	No unauthorized processing; controlled error.
Backend times out	Local submission remains consistent; failure is recorded without exposing secrets.
Response is malformed	Result is rejected rather than silently applied.

Implementation Status:

The uploaded application contains Deluge custom functions and caller definitions. It contains no evidenced Catalyst endpoint, connection or invokeurl integration. The external-service architecture and test cases here are an exploration plan, not a completed deployment.

Result:

The custom-function approach was examined through report-count refresh, renewal reminders and report navigation in the Gym Membership application. A Catalyst extension procedure and interface contract were prepared. Catalyst deployment, authenticated invocation and live test results remain to be demonstrated.